

PROJECTS & EXPERIENCE

Amazon Web Services, Braket

SDE Intern

June 2026 — September 2026

- Modeled graph compilation infrastructure lowering high level intermediate representations (IR) into native execution
- Formulated topological routing constraints and instruction scheduling passes to satisfy physical hardware
- Modeled low latency runtime interfaces to support dynamic mid circuit execution and classical control flow feedback loops for real time workload mitigation.
- Received official AWS return endorsement for systems performance engineering and cross stack hardware software codesign.

Quantum Codesign Lab

Niu Group, UCSB Undergraduate Researcher

June 2025 — Present

- Modeled optical matrix vector multiplication (MVM) execution pipelines and scattering matrix operations for synthetic optical neural network architectures within [\(Green Machine architectures\)](#)
- Trained and benchmarked Recurrent Neural Network (GRU) decoders for continuous time series inference on noisy streaming data

FX2Accel & AcelSim: Compiler IR & Architectural Simulation Framework

Language: Python

January 2025 — Present

- Built an ML compilation and simulation pipeline coupling PyTorch FX graph lowering passes with a cycle accurate hardware
- Implemented ShapeProp, dead code elimination (DCE), and operator fusion passes to minimize instruction counts and eliminate intermediate DRAM roundtrips.
- Designed a memory planner for on chip SRAM buffer reuse to resolve hardware constraints and optimize memory footprints.
- Modeled a 5 instruction spatial ISA with configurable latency, tracking memory traffic and classifying workload bottlenecks

KernelForge

Language: Python, C++, CUDA

March 2026 – May 2026

- Designed a Triton Inspired Domain Specific Language (DSL) and kernel code generation compiler for high throughput deep learning primitives (fused GEMM and elementwise operations).
- Implemented tile level block scheduling and pointer arithmetic passes to ensure memory coalescing and minimize DRAM roundtrips across GPU global memory.

EDUCATION

BS/MS ECE | University of California, Santa Barbara

Emphasis in Computer Architecture, ML Compilers, and AI Hardware Acceleration.

August 2023 — December 2027

COURSE WORK

Embedded Systems (ECE153B)	Digital Design (ECE152)	Matrix Analysis (ECE 210)
Optoelectronic Physics (ECE162)	Signal Processing (ECE 130)	Waveguide Physics (ECE 144)
Optimizations & ML (ECE 133)	Probability & Stats (ECE 139)	Fiber Optics (ECE 135)

SKILLS & LEADERSHIP

Languages : C++, C, Python, Java, Arduino, LATEX

Tools : Unix, Git/Github, LangChain, Tavily, OpenAI, Google Cloud